HARINI A K

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Career Objective

Final-year Computer Science student with experience in Java, Python, full-stack development, and AI-powered applications. Skilled in building scalable backend systems, databases, and modern web applications. Seeking an opportunity to contribute to innovative software engineering projects while continuously learning emerging technologies.

Education

Bachelor of Technology in Computer Science Engineering, **2023 – 2027**
Amrita Vishwa Vidyapeetham, Coimbatore Coimbatore
CGPA: 6.51

Skills

Programming Languages
Java, Python, C

Frontend
HTML, CSS, React.js, Next.js, TypeScript

Backend
Node.js, Express.js, FastAPI, Redis, Celery

Databases
MySQL, PostgreSQL, Supabase

Tools
Git, Github, VSCode

Projects

Adaptive Traffic Signal Control Using Fog Computing & RL, *Python, PPO, ANN, SUMO, TraCI*

- Designed an intelligent traffic signal control system integrating Artificial Neural Network (ANN) and Proximal Policy Optimization (PPO) for real-time adaptive decision-making.
- Implemented and evaluated the model in SUMO simulator using TraCI, achieving over 99% reduction in vehicle waiting time compared to fixed-time signal control

AI-Enabled Crop Disease Diagnosis & Severity Detection System, *FastAPI, PostgreSQL, Redis, Celery, PyTorch*

- Developed REST APIs using FastAPI for AI model inference and severity classification (~96% accuracy).
- Designed backend with PostgreSQL and implemented asynchronous processing using Celery & Redis.
- Built scalable architecture to handle multiple prediction requests efficiently.

Golf Charity Subscription Platform, *Full Stack*

- Built a full-stack subscription platform with Next.js and Supabase featuring golf score tracking, a monthly charity draw system, and an admin panel for end-to-end management
- Designed a 5-table PostgreSQL schema and implemented core logic including a score replacement algorithm, random draw matching, and session-based route protection

• Live: https://golf-charity-platform-1gmj.vercel.app/

Project Tracker, *React, TypeScript*

- Built a multi-view project management UI in React and TypeScript with custom drag-and-drop (native pointer events, no libraries), virtual scrolling for 500+ tasks, and instant Kanban/List/Gantt view switching over shared state
- Implemented URL-synced filters for shareable views, simulated real-time collaboration indicators via mock WebSocket, and achieved a Lighthouse performance score of 85+

• Live: https://project-tracker-owa1.vercel.app/

HACKATHONS

Participant – ByteVerse 7.0 Hackathon by NIT Patna, **04/2025 – 04/2025**

Python, Streamlit, OpenAI API, HTML

- Developed an AI-powered web app promoting mental wellness through conversational AI, mood tracking, and interactive games within a limited time frame as part of a collaborative team effort.
- Contributed to front-end/back-end, enhancing problem-solving and time management skills.

Extra Curricular Activities

- Actively participated in NSS for 2 years, contributing to community service and social welfare programs.
- Elected Class Representative (3rd Year), coordinating communication between faculty and 60+ students.